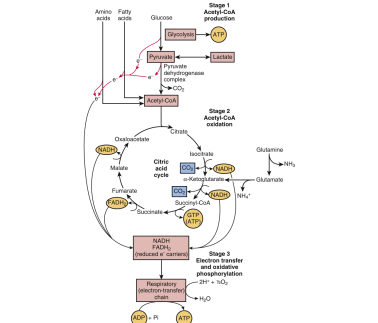
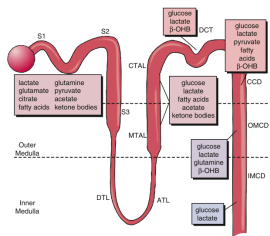


Fig. 5.1 Renal metabolism may influence renal physiology, health, and disease through energy production or regulatory mechanisms not directly related to energy production. NO, Nitric oxide; ROS, reactive oxygen species; TCA cycle, tricarboxylic acid cycle.

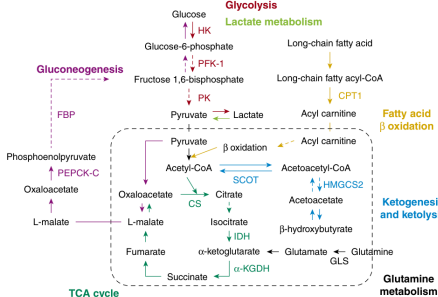
Fig_5_1



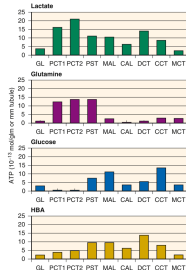
Fig_5_4



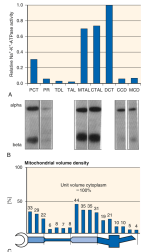
Fig_5_7



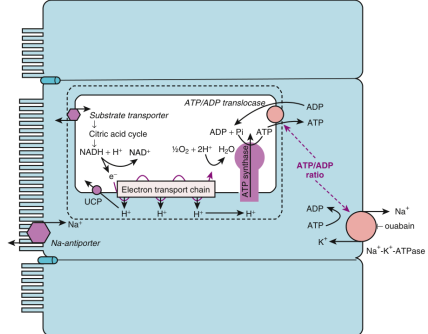
Fig_5_2



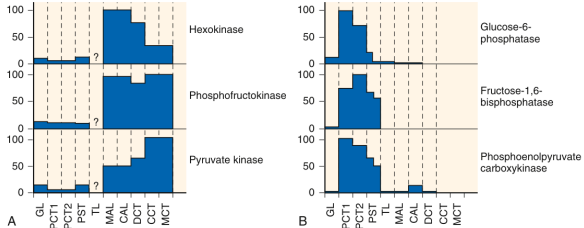
Fig_5_5



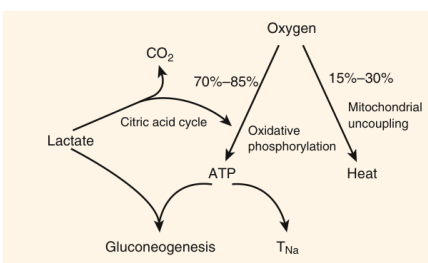
Fig_5_8



Fig_5_3



Fig_5_6



Fig_5_9

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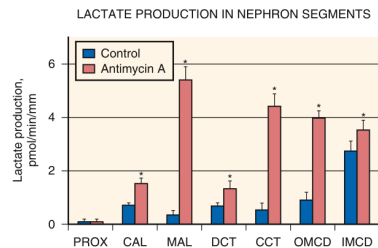


Fig. 5.10 Lactate production by rat nephron segments under control conditions and in the presence of antimycin A. *CAL*, Cortical ascending limb; *CCT*, cortical collecting tubule; *DCT*, distal convoluted tubule; *IMCD*, inner medullary collecting duct; *MAL*, medullary thick ascending limb; *OMCD*, outer medullary collecting duct; *PROX*, proximal tubule. (From Bagnasco S, Good D, Balaban R, et al. Lactate production in isolated segments of the rat nephron. *Am J Physiol*. 1986;248:F522-F526.)

Fig_5_10

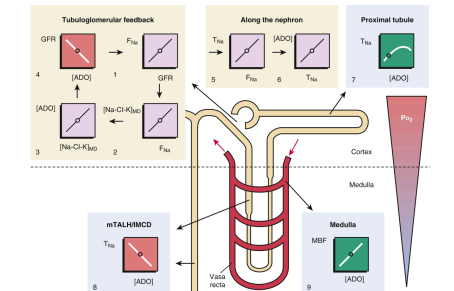


Fig. S13 Role of extracellular adenosine (ADO) in protecting the renal medulla from hypoxia. The line plots indicate the relationships between the given parameters. Small circles on these lines indicate absolute physiologic conditions. 1. A rise in glomerular filtration rate (GFR) increases the N_a^+ load (F_{Na}) to the tubular system in cortex and medulla. 2. This rise in F_{Na} increases the salt concentration served by the medullary thick ascending loop back loop to provide a basis for an osclating system. 3. F_{Na} determines N_a^+ transport (T_{Na}) and O_2 consumption in every nephron segment, and thus oscillations in F_{Na} may help protect the medulla. 4. A rise in T_{Na} increases ADO along the nephron. 7. In the cortical proximal tubule, ADO stimulates T_{Na} and thus lowers the N_a^+ load to segments residing in ADO. 8. In contrast, ADO inhibits transport in the thick ascending loop, reducing the salt ascending the second limb and thereby collecting duct (ACD). 9. This inhibition decreases medullary blood flow (MBF), which reduces the ADO concentration in the medulla. 10. ADO stimulates medullary blood flow (MBF), which increases ADO and kidney function. *Physiol Rev* 2006;86(919-946).

Fig_5_13

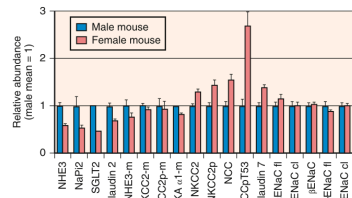


Fig. 5.11 Profile of renal sodium transporter protein abundance in female C57BL/6 mice expressed relative to abundance in male rats cotransporter isoform 2 ($\text{NaP}2\alpha$). Cortical and medullary (-m) samples were detected by immunoblot: Na⁺/H⁺ exchanger isoform 3 (*NHE3*), Na⁺-phosphate cotransporter isoform 2 (*NaP2*), Na⁺-glucose cotransporter (SGLT2), claudin 2, Na⁺-K⁺-ATPase cotransporter isoform 2 (*KVCC2*) and activated calyculin A (*CALY*). The expression profile of each transporter was determined from densitometric analysis of the blots obtained in triplicate. Data are shown as mean $\pm$ SEM. * indicates significant difference between groups ($p < 0.05$, Student's *t*-test). Scale bar = 10 kDa. Adapted from Verma LS, Garg V, et al. (2008) Differential expression of renal sodium transporters in male and female C57BL/6 mice. *J Am Soc Nephrol* 19:228-235.

Fig_5_11

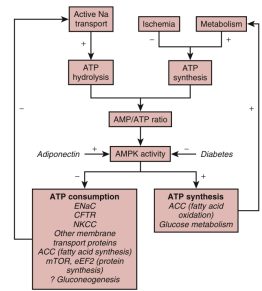


Fig. 5.14 Proposed role of adenosine monophosphate-activated protein kinase (AMPK) in the kidney in coupling catabolic pathways requiring ATP hydrolysis (primarily sodium transport) with metabolic pathways leading to ATP synthesis (primarily fatty acid and glucose oxidation). +, Activating pathway; -, inhibitory pathway. ACC, acetyl-CoA carboxylase. (From Hallows KR, Mount PF, Pastor-Dorier M, Power DA. Role of the energy sensor AMP-activated kinase in renal physiology and disease. *Am J Physiol Renal Physiol.* 2010;298:F1067-F1077.)

Fig_5_14

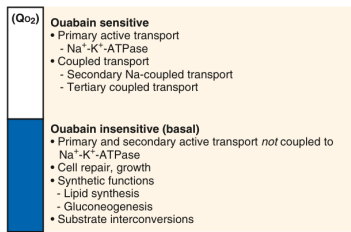


Fig. 5.12 Ouabain-sensitive and -insensitive oxygen consumption in the kidney. A large fraction of renal epithelial oxygen consumption (Q_{O_2}) in renal cells is sensitive to the Na^+K^+ -ATPase-specific inhibitor ouabain, and this Q_{O_2} drives primary active transport and transport coupled with sodium pump activity. The fraction of renal oxygen consumption that does not change in the presence of ouabain is, by definition, independent of Na^+K^+ -ATPase activity in the cell and is roughly equivalent to the basal Q_{O_2} , which fuels transport not coupled to sodium gradients, cell repair and growth, biosynthesis, and substrate interconversions.

Fig_5_12

Process	Direct Product	Final ATP
ATP Yield from Complete Oxidation of Glucose		
Glycolysis	2 NADH (cytosol)	5*
	2 ATP	2
Pyruvate oxidation (2 per glucose)	2 NADH (mitochondrial matrix)	5
Acetyl-coenzyme A oxidation in citric acid cycle (2 per glucose)	6 NADH (mitochondrial matrix)	18
	2 FADH ₂	4
Total yield per glucose		30
ATP Yield from Glycolysis of Glucose		
Glycolysis	2 ATP, 2 NADH	2

a tissue-specific manner. UCPs create a proton leak that dissipates the proton gradient available to drive oxidative phosphorylation (see Fig. 5.3). It has been reported that UCP-2 is expressed in the renal proximal tubule and thick ascending limb (TAL) (not in glomerulus or the distal nephron) and that its expression is elevated in kidneys of diabetic rats.⁷

Table_5_1